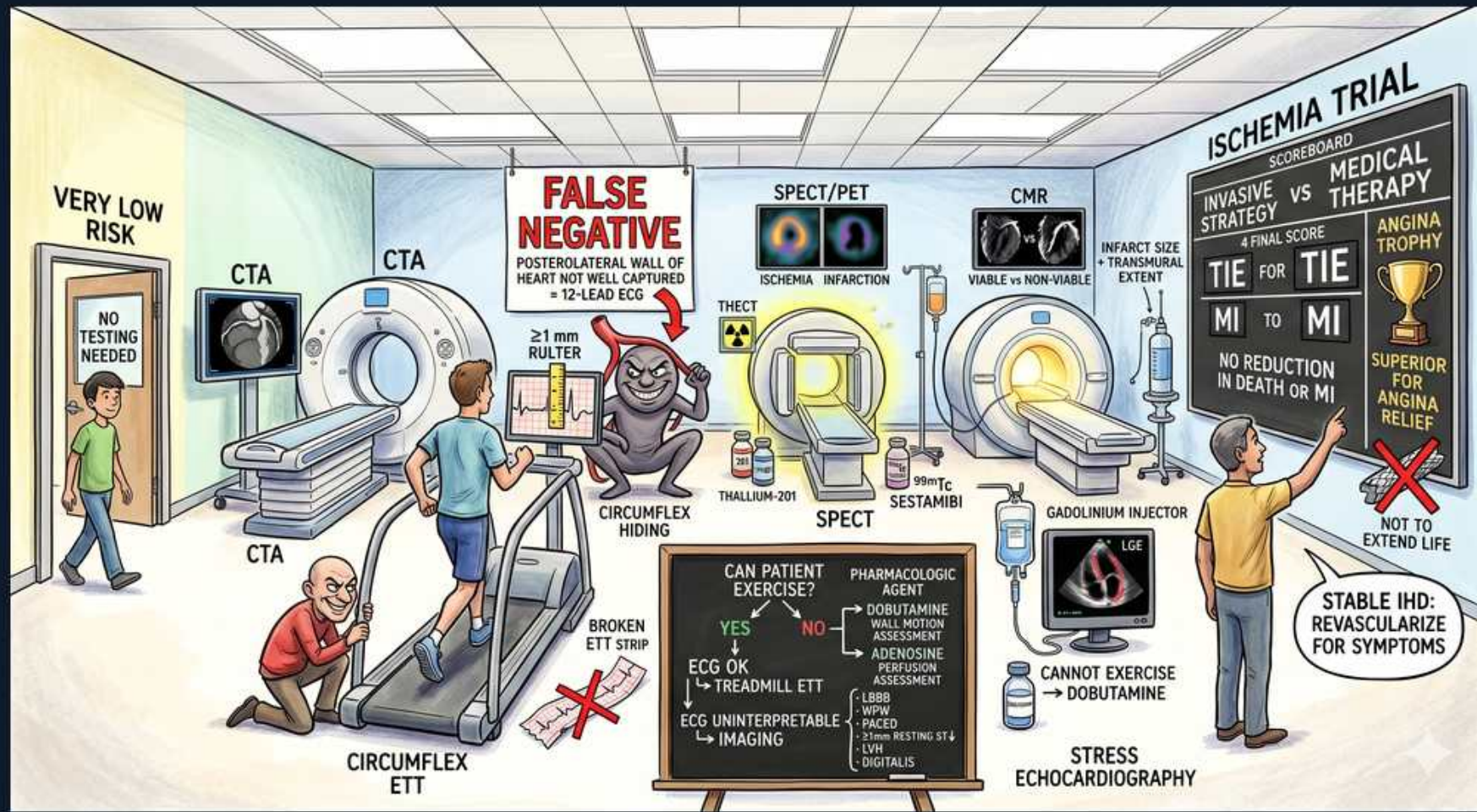


Stable IHD: Choosing the Non-Invasive Test & the ISCHEMIA Lesson



1. Very low risk — no testing

WHAT YOU SEE

Green man, NO TESTING NEEDED sign

WHAT IT ENCODES

In patients with very low pre-test probability of CAD, no further testing is needed — testing only raises false positives. Board point: do not order stress tests reflexively in low-risk chest pain; the post-test probability barely changes and a positive result is more likely false.

BOARD TRAP

Ordering CTA or stress in low-PTP patients — the discriminator is pre-test probability, not symptom presence.

2. Coronary CTA

WHAT YOU SEE

CTA scanner with coronary tree images

WHAT IT ENCODES

Coronary CT angiography is the preferred test in low-to-intermediate PTP because of its high negative predictive value — it rules OUT obstructive CAD. Excellent anatomic test; high sensitivity. Limited by coronary calcium and need for low/regular heart rate.

BOARD TRAP

Choosing CTA to assess ischemia/viability — CTA shows anatomy, not function; it cannot tell you if a stenosis is flow-limiting.

3. Exercise ECG false negative — circumflex

WHAT YOU SEE

FALSE NEGATIVE banner, gorilla 'circumflex hiding'

WHAT IT ENCODES

The 12-lead ECG poorly captures the posterolateral wall, so circumflex-territory ischemia is a classic false-negative on exercise ECG. Standard ECG has a posterolateral blind spot. Board point: a normal exercise ECG does not exclude LCx disease.

BOARD TRAP

Calling exercise ECG definitively negative — the circumflex blind spot is the discriminator behind a missed posterolateral MI.

4. Exercise ECG ≥1mm flat/downsloping ST

WHAT YOU SEE

≥1mm RULER, broken ETT strip

WHAT IT ENCODES

A positive exercise ECG is ≥1mm horizontal or downsloping ST depression. Upsloping ST depression does NOT count as positive. Used in patients who can exercise and have an interpretable baseline ECG.

BOARD TRAP

Upsloping ST depression labeled positive — only flat/downsloping ≥1mm qualifies.

5. SPECT / PET perfusion

WHAT YOU SEE

SPECT/PET, thallium-201 & sestamibi vials

WHAT IT ENCODES

Nuclear perfusion (SPECT thallium-201 or technetium-99m sestamibi, PET) distinguishes ischemia (reversible defect) from infarction (fixed defect). Reversible = viable ischemic tissue; fixed = scar. Adds functional/perfusion data the ECG cannot.

BOARD TRAP

Reading a fixed defect as active ischemia — fixed = scar/non-viable, reversible defect = inducible ischemia.

6. Cardiac MRI — viability & transmural extent

WHAT YOU SEE

CMR scanner, viable vs non-viable, infarct size = transmural extent

WHAT IT ENCODES

CMR with late gadolinium enhancement quantifies infarct size by transmural extent of scar. <50% transmural late enhancement predicts functional recovery after revascularization (viable); >50% is non-viable. Best test for viability assessment.

BOARD TRAP

Revascularizing a wall with >50% transmural LGE expecting recovery — that segment is scar and will not improve.

7. Can patient exercise? — modality choice

WHAT YOU SEE

Chalkboard YES/NO branch, treadmill vs pharmacologic

WHAT IT ENCODES

First decision in stress testing: can the patient exercise? If yes and ECG interpretable → exercise ECG. If ECG uninterpretable (LBBB, paced, baseline ST) → add imaging. If cannot exercise → pharmacologic stress.

BOARD TRAP

Doing exercise ECG with an uninterpretable baseline (LBBB/paced) — you must add imaging or use a perfusion/pharmacologic protocol.

8. Pharmacologic stress agents

WHAT YOU SEE

Adenosine/regadenoson perfusion vs dobutamine

WHAT IT ENCODES

If a patient cannot exercise, use pharmacologic stress: vasodilators (adenosine/regadenoson/dipyridamole) for perfusion imaging, or dobutamine for stress echo. Vasodilators induce coronary steal; dobutamine increases myocardial demand.

BOARD TRAP

Giving adenosine to a severe-asthma/AV-block patient — bronchospasm/heart block; dobutamine is the alternative.

9. Stress echocardiography

WHAT YOU SEE

Stress echo board, dobutamine for those who cannot exercise

WHAT IT ENCODES

Dobutamine stress echo detects inducible wall-motion abnormalities and assesses contractile reserve/viability. Used when patients cannot exercise. A new regional wall-motion abnormality with stress indicates inducible ischemia.

BOARD TRAP

Expecting perfusion data from stress echo — echo shows wall motion, not perfusion; choose nuclear if perfusion is the question.

10. ISCHEMIA trial — invasive vs medical

WHAT YOU SEE

ISCHEMIA TRIAL scoreboard, TIE for TIE

WHAT IT ENCODES

The ISCHEMIA trial showed that in stable CAD, an initial invasive strategy gave NO reduction in death or MI versus optimal medical therapy. It tied for hard endpoints. This reframed revascularization in stable disease.

BOARD TRAP

Claiming early angiography reduces mortality in stable IHD — ISCHEMIA showed no death/MI benefit over OMT.

11. Revascularize for symptoms, not survival

WHAT YOU SEE

Angina trophy 'superior for angina relief', STABLE IHD box

WHAT IT ENCODES

Per ISCHEMIA, revascularization in stable IHD is superior for ANGINA RELIEF and quality of life but does NOT extend life. Board point: in stable IHD revascularize for refractory symptoms, not to prolong survival.

BOARD TRAP

PCI 'to extend life' in stable IHD — the indication is symptom control; left main / high-risk anatomy are the survival-benefit exceptions.
